Problem 4: Let's consider an infinite sequence of digits constructed of ascending powers of 10 written one after another. Here is the sequence: 110100100 0...
You are to find out what digit is located a the definite position of the sequence. (Timus. 1209)

Solution:

1-1	2-1	3-1	4-1	5-1	6-1	→ Zeroes present
1	10	100	1000	10000	100000	....
↓	↓	↓	↓	↓	↓	
1	2	4	7	11	16	

If we observe the position of the digits with '1' in their placeholder, they are actually the values of (triangular number +1). For example, in case of 7, 1 has three zeroes (0) with it and total number of digits is 4. And the summation of from 1 to 3 (1 is excluded, only number of 0s are counted) is $\frac{3(3+1)}{2} = 6$ and $6+1 = 7$ where the digit 1 is.

$\therefore$ 1s are located at (triangular number + 1) th places or, $\left(\frac{n(n+1)}{2}+1\right)$ th digits are 1s and here n starts from 0.

Prerequisite:

1. Same as problem no. 1.

Discussion / Furthermore procedures:

Tri-angular numbers are integers and non-negative.

Let, these triangular number be denoted as T_i.

$$\therefore T_i = \frac{i(i+1)}{2} \rightarrow T_i + 1 = \frac{i(i+1)}{2} + 1$$

Here, (T_i+1) th positioned digit is '1'. Let, $(T_i+1) = z$.

$$\therefore z = \frac{i(i+1)}{2} + 1 \rightarrow z = \frac{i^2+i+2}{2}$$

$$\rightarrow 2z = i^2 + i + 2 \rightarrow i^2 + i = 2(z-1)$$

$$\rightarrow 1.i^2 + 1.i^1 + (-2(z-1))\, i^0 = 0 \quad \ldots (1)$$

From equation (i), we can apply the formula of quadric equation for 'i',

$$i = \frac{-(1) \pm \sqrt{(1)^2 - 4 \times (1) \times (-2(z-1))}}{2.(1)}$$

$$= \frac{-1 \pm \sqrt{1 + 8(z-1)}}{2}$$

$$= \frac{\sqrt{8z-7} - 1}{2}, \frac{-1 - \sqrt{8z-7}}{2} \quad \ldots (a)$$

Some values of z or $(T_i + 1)$ are,

$z = \{1, 2, 4, 7, 11, 16, \ldots\}$

if we plugin the values, we will find that, $\sqrt{8z-7}$ produces integer values. So, $\sqrt{8z-7}$ has to be always a square number.

We are dicarding the (-) root because tri-angular numbers are by definition the summation of sequenced positive integers. Also, the given problem is a positive number.